

AAPL Strategy Backtesting Report

Technical Indicators & Strategy Backtesting with Alpaca

Apple Inc. (AAPL)

Backtest Assumptions

- Initial capital: \$100,000.
- Long-only portfolio construction.
- No leverage and no short selling.
- Whole-share position sizing.
- Signals generated from daily OHLCV data are executed at the next day open to reduce look-ahead bias.

Executive Summary

Conclusion: For AAPL, Buy & Hold produced the highest absolute return, but the best risk-adjusted active strategy was the **Custom Volume-Confirmed Trend Pullback**. It delivered 68.06% total return with a Sharpe ratio of 1.05, a Sortino ratio of 0.96, and maximum drawdown of -9.86%. This makes it the strongest active strategy when the assignment objective is risk-adjusted performance rather than raw return.

Best Absolute Return	Best Sharpe Ratio	Lowest Drawdown	Highest Win Rate
Buy & Hold 125.85%	Custom: Volume-Confirmed Trend Pullback 1.05	Mean Reversion -6.26%	Mean Reversion 87.50%

Performance Comparison

We compare the benchmark and three active strategies. Values are calculated from daily portfolio values. Return, CAGR, volatility, maximum drawdown, win rate, and exposure are shown as percentages.

Strategy	Ending Value	Total Return	CAGR	Volatility	Sharpe	Sortino	Max Drawdown	Win Rate	Round Trips	Exposure
Buy & Hold	\$225,849.36	125.85%	17.65%	27.51%	0.72	1.04	-33.32%	N/A	0	100.00%
Trend Following	\$104,160.47	4.16%	0.82%	6.54%	0.16	0.07	-12.20%	52.63%	19	11.64%
Mean Reversion	\$145,260.92	45.26%	7.73%	10.34%	0.77	0.41	-6.26%	87.50%	8	7.76%
Custom: Volume-Confirmed Trend Pullback	\$168,057.10	68.06%	10.91%	10.43%	1.05	0.96	-9.86%	56.36%	55	24.86%

Main Findings

- Buy & Hold dominated on raw wealth creation: the portfolio finished at \$225,849.36 for a total return of 125.85% and CAGR of 17.65%. The drawback is that it carried full market exposure and experienced a much deeper maximum drawdown of -33.32%.
- The Custom Volume-Confirmed Trend Pullback was the best active strategy on a risk-adjusted basis. Its Sharpe ratio of 1.05 exceeded Buy & Hold at 0.72 while using only 24.86% exposure. This suggests the strategy captured meaningful upside while spending substantial time out of the market.
- Mean Reversion was the strongest capital-preservation strategy. It had the lowest maximum drawdown at -6.26% and the highest win rate at 87.50%. However, it generated less total return than the custom strategy.
- Trend Following underperformed materially. It finished with only 4.16% total return and a Sharpe ratio of 0.16. The strict MACD, ADX, and moving-average filters kept the strategy out of many profitable AAPL moves and led to weak realized performance.

Portfolio-Level Results

Equity Curve Comparison

The equity curve shows the primary tradeoff. Buy & Hold accumulated the most wealth because AAPL appreciated significantly over the sample period. However, the active strategies reduced time in the market, producing smoother but lower equity curves. The custom strategy was the strongest active curve, finishing well above both Trend Following and Mean Reversion.

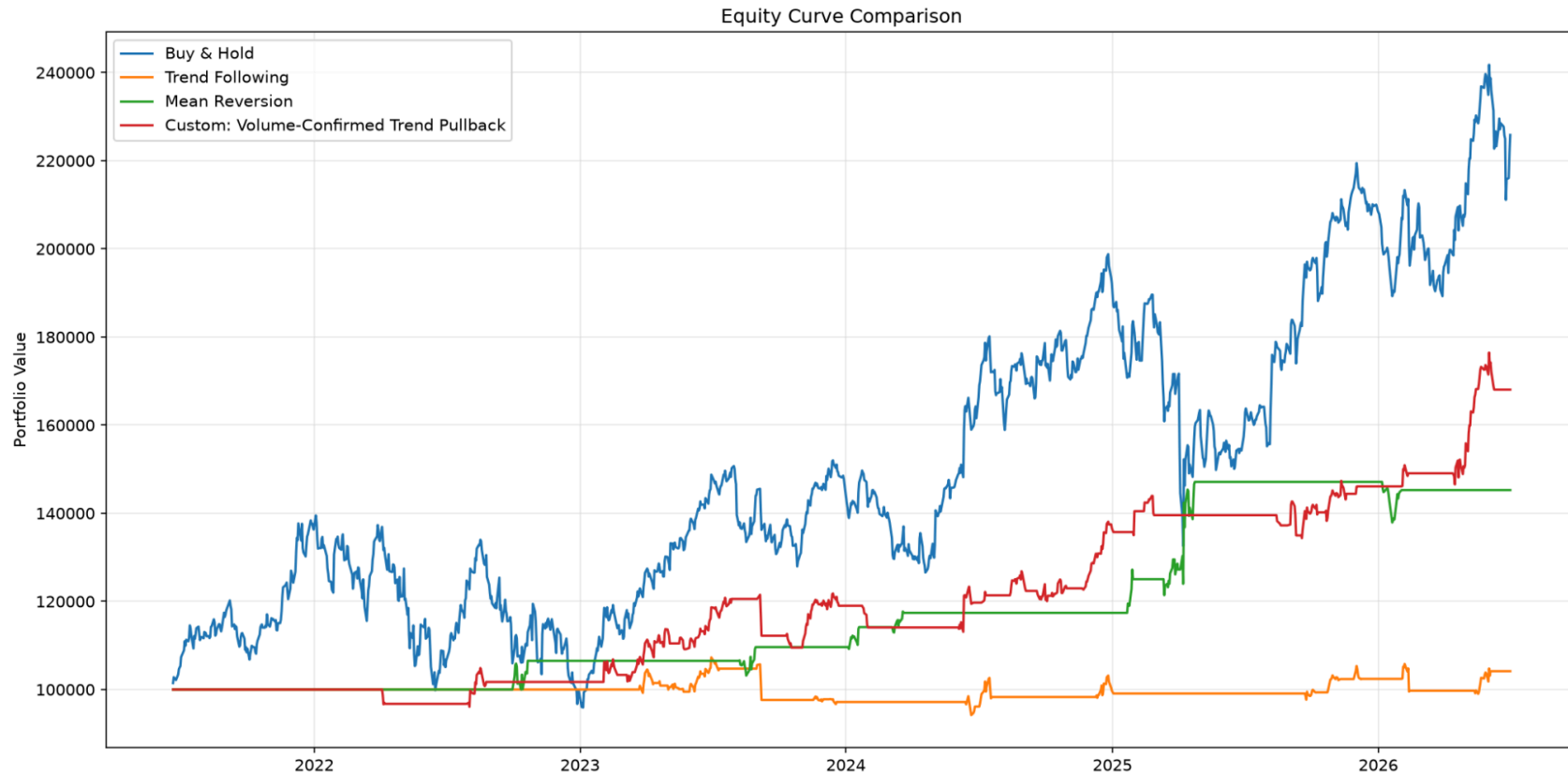


Figure 1. Equity curve comparison across Buy & Hold and active strategies.

Drawdown Comparison

The drawdown chart highlights why risk-adjusted metrics matter. Buy & Hold suffered the deepest drawdowns, including a drawdown greater than 30%. Mean Reversion had the smallest drawdown profile because it was rarely exposed and entered only after oversold conditions. The custom strategy accepted slightly higher drawdown than Mean Reversion, but compensated with substantially higher return and the best Sharpe ratio among all strategies.

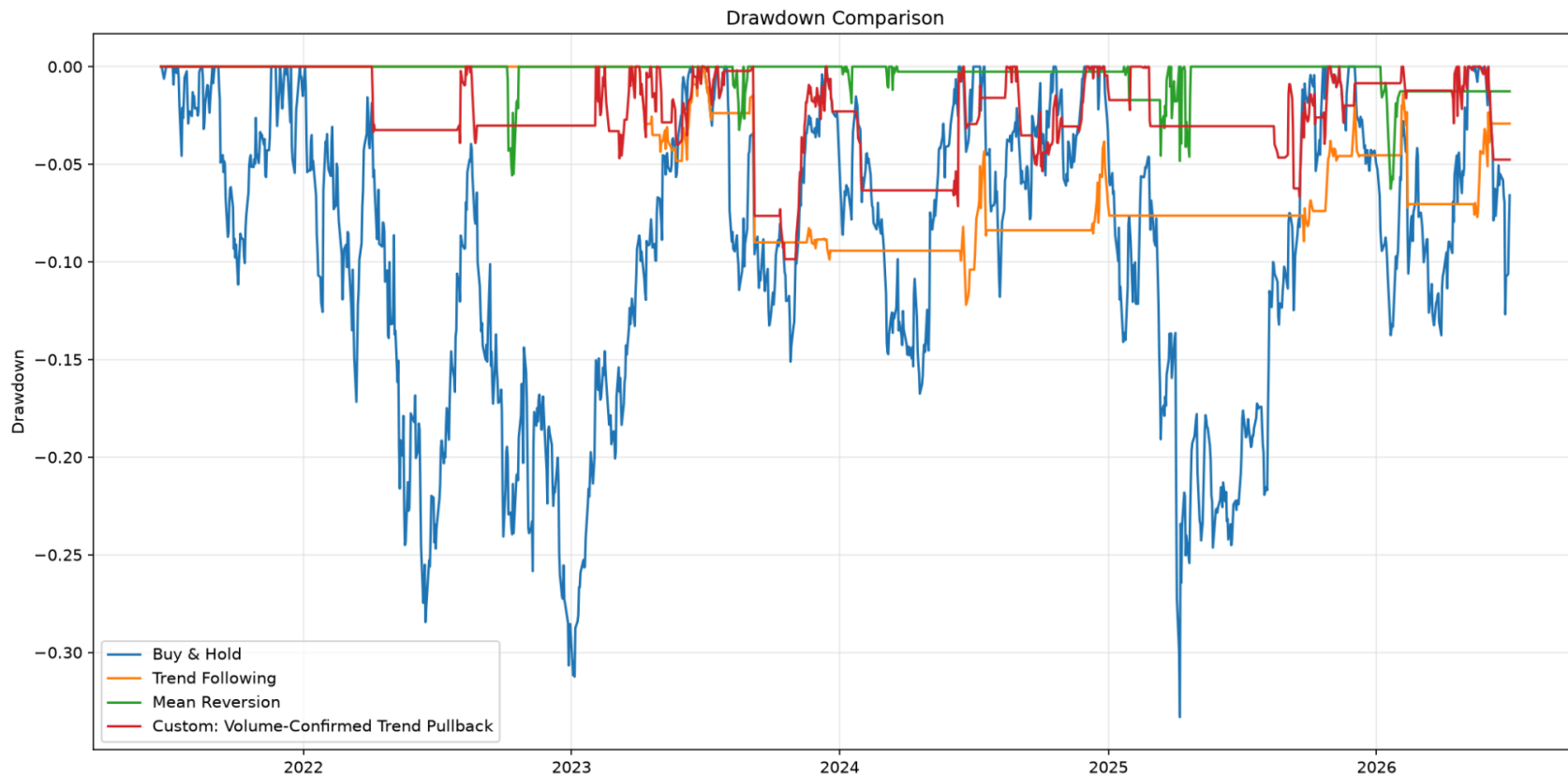


Figure 2. Drawdown comparison. Negative values show percentage decline from each strategy equity high-water mark.

Strategy Review

Strategy 1: Trend Following

Rules: Buy when MACD is above the MACD signal, ADX is above 25, SMA(50) is above SMA(200), and close is above SMA(50). Sell when MACD falls below signal, close falls below SMA(50), or ADX falls below 18.

The Trend Following strategy was too conservative for AAPL in this sample. It was exposed only 11.64% of the time and produced only 4.16% total return. The chart shows that many entries occurred after AAPL had already advanced, while exits often came after short-term weakening. The ADX and moving-average confirmation requirements reduced participation in the main upside trend.

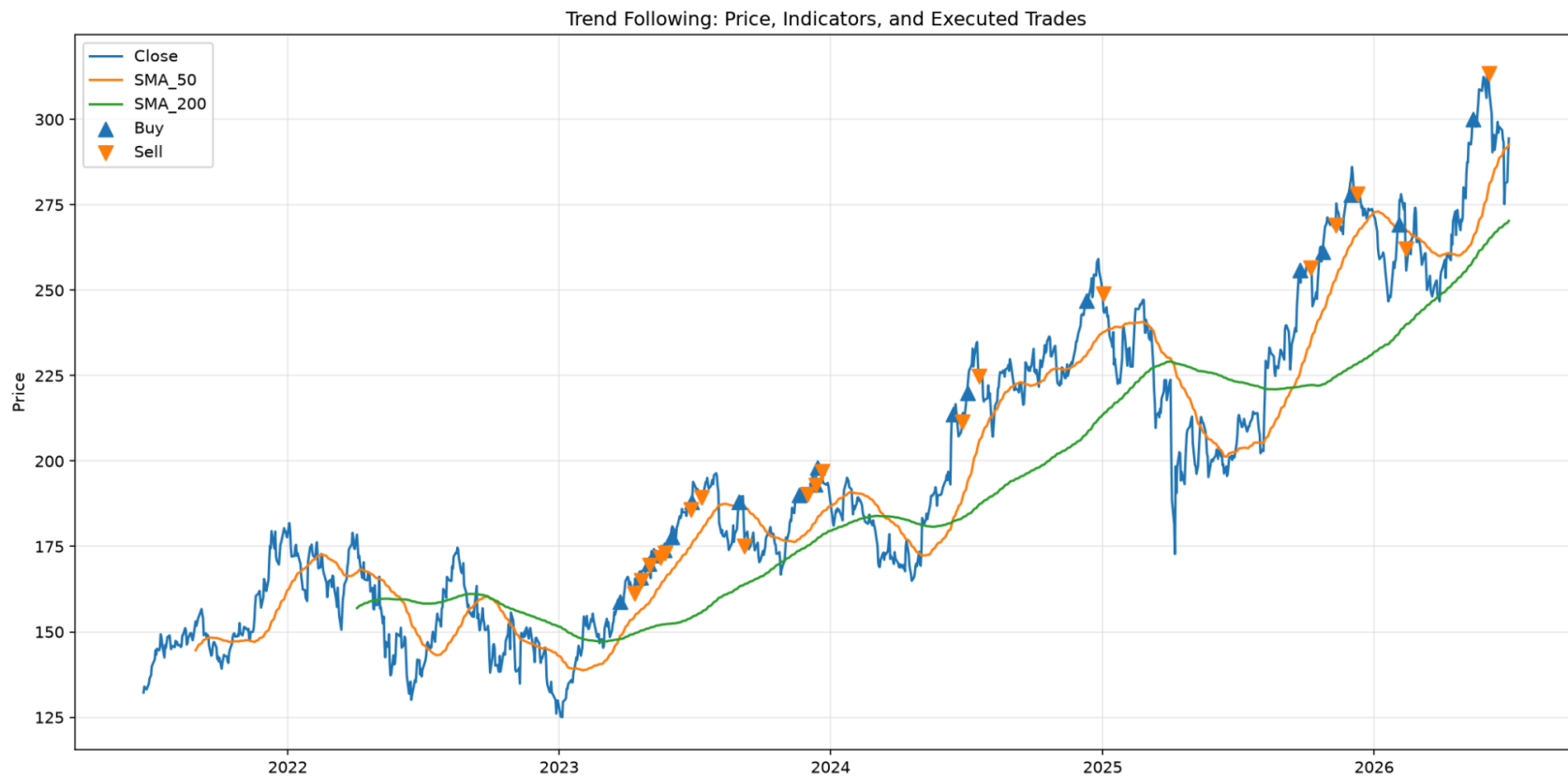


Figure 3. Trend Following signals overlaid on AAPL price and moving averages.

Strategy 2: Mean Reversion

Rules: Buy when RSI(14) is below 30 and close is below the lower Bollinger Band. Sell when price mean-reverts above the middle Bollinger Band, RSI rises above 55, or overbought conditions appear.

Mean Reversion performed well defensively. It produced 45.26% total return with only 7.76% exposure and the smallest maximum drawdown of -6.26%. Its 87.50% win rate indicates that oversold AAPL entries often bounced. The limitation is opportunity cost: the strategy frequently stayed in cash during sustained uptrends, so it could not match Buy & Hold or the custom strategy on total return.

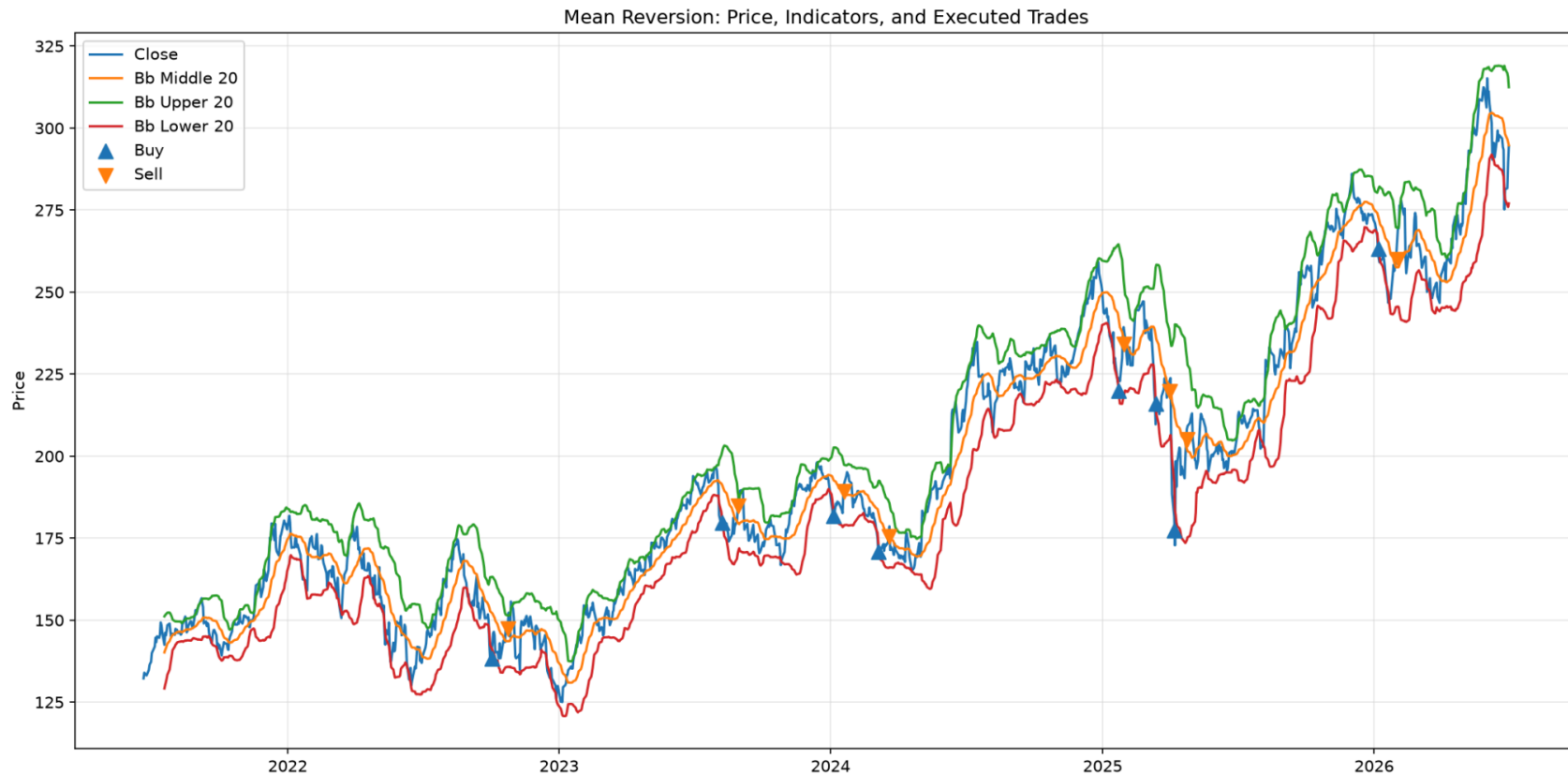


Figure 4. Mean Reversion signals using RSI and Bollinger Band context.

Strategy 3: Custom Volume-Confirmed Trend Pullback

Rules: Buy when AAPL is above SMA(200), Momentum(10) is positive, close is above the Bollinger middle band, OBV is above its 20-day average, and CMF(20) is positive. Sell when close falls below SMA(50), Momentum(10) turns negative, CMF(20) turns negative, or RSI exceeds 75.

This was the most balanced active strategy. It returned 68.06% with a Sharpe ratio of 1.05, a Sortino ratio of 0.96, and maximum drawdown of -9.86%. The strategy traded more often than the others, with 55 round trips, but its volume confirmation helped avoid weak trend signals. It did not beat Buy & Hold on raw return, but it achieved better risk-adjusted performance with substantially less exposure.

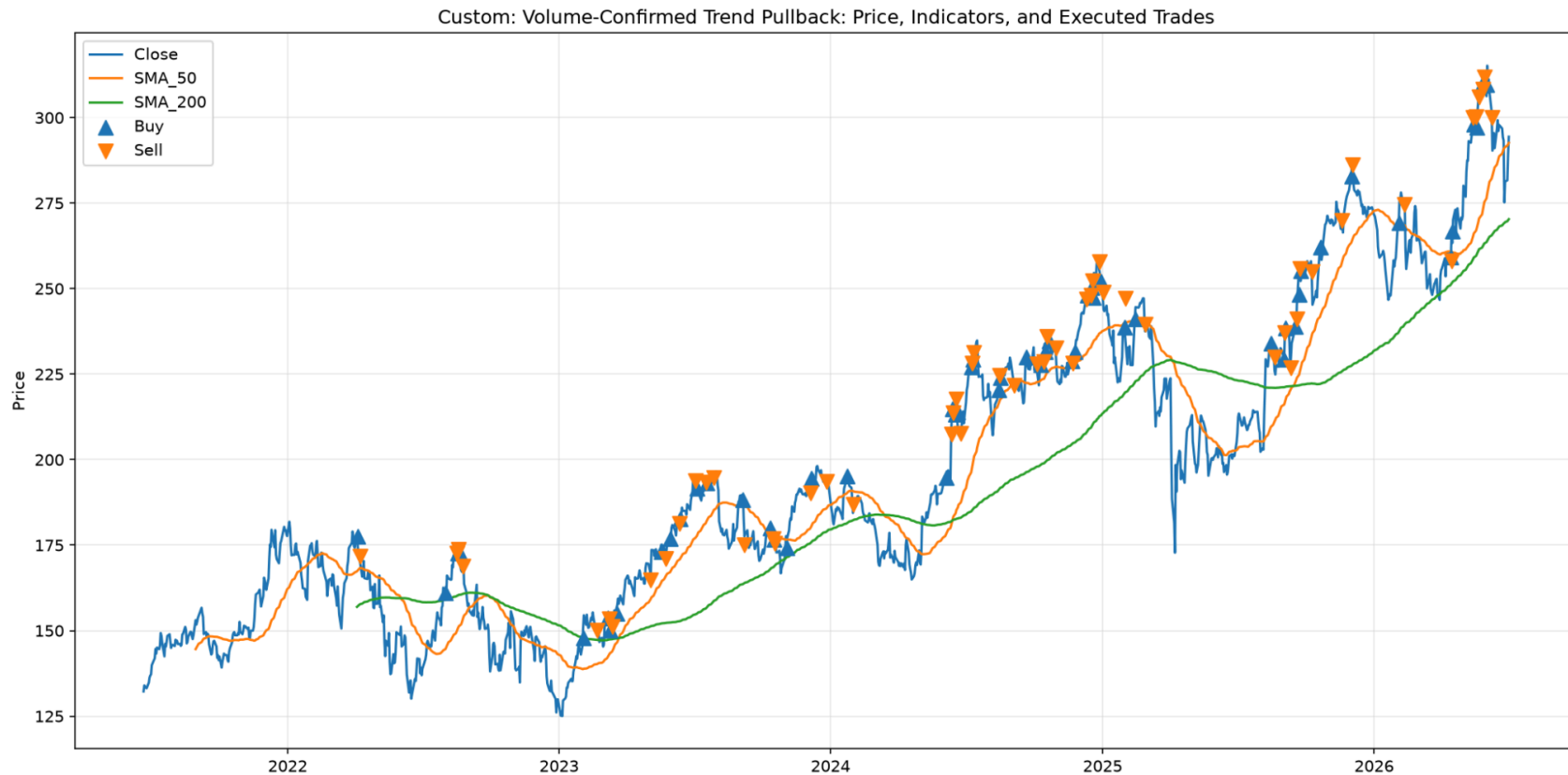


Figure 5. Custom strategy signals using trend, momentum, volatility/location, and volume confirmation.

Recommendation for AAPL

For AAPL, if the objective is maximum ending wealth and the investor can tolerate large drawdowns, Buy & Hold is the best choice. It benefited from AAPL's strong long-term appreciation and finished far ahead of the active strategies in dollar terms.

However, based on a risk-adjusted basis, our Custom Volume-Confirmed Trend Pullback is the objective winning strategy. It generated the highest Sharpe ratio in the table and the strongest active-strategy ending value while maintaining much lower exposure than Buy & Hold.

Mean Reversion is the preferred defensive alternative. It had the smallest drawdown and highest win rate, making it attractive for a capital-preservation style. The tradeoff is that it left a large amount of upside on the table because it was active for less than 8% of the backtest.

Trend Following should not be selected in its current form. Its strict filters likely caused late entries, premature exits, and too much time in cash.